

Filip Rehburg

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Current Education

INSTITUTE OF LOGIC, LANGUAGE AND COMPUTATION, UvA

Amsterdam, NL

MSc in Logic, September 2025 -
Computation Track,
Member of the Diversity Committee

Prior Education

MAASTRICHT UNIVERSITY

Maastricht, NL

BSc (Honors) in Data Science and Artificial Intelligence, June 2023
Thesis: Electrocardiographic Imaging of Synthesized Human Data Using Deep Learning
Advisors: Prof. Ralf Peeters and Dr. Joël Karel

CITY UNIVERSITY OF HONG KONG

Kowloon, Hong Kong SAR

Exchange semester at Department of Data Science

Research Interests

Reasoning, Neuro-Symbolic AI, Deep Learning, Graph Neural Networks, Knowledge Representation, Modal Logic

Relevant Courses

UvA: [52041DEL6Y](#) Deep Learning 1 8/10
UM: [KEN2520](#) Machine Learning 9/10
UM: [KEN2110](#) Graph Theory 9/10

Experience

12/2023 - 12/2024 **Research Fellow Medical Data Science** *Berlin Institute of Health at Charité, Berlin, DE*
Training classification models to assist diagnosing rare diseases in automated clinical applications using data with strong class imbalance. Codeveloped a framework for storing and exchanging rare disease data to facilitate cross-location analysis. Contributed to the phenopackets and gpsea projects. Authored one and contributed to five publications.

8/2023 **Research Intern Medical Data Science** *Berlin Institute of Health at Charité, Berlin, DE*

9/2021 - 7/2022 **Research Assistant (MaRBLe)** *Ludii Research Group, DACS, UM, Maastricht, NL*
Developed a code completion software for the bespoke Ludii programming language using classical NLP approaches and achieved high accuracy with minimal latency.

Awards and Honors

4/15 Successfully completed MaRBLe Honors program
15/60 Accepted into MaRBLe Honors program: year-long research project on code completion
3/100+ Selected for one out of three spots for an exchange semester in Hong Kong

Research and Publications

Journal

- [1] Graefe ASL, **Rehburg F**, Alkarkoukly S, ..., & Beyan O. RareLink: scalable REDCap-based framework for rare disease interoperability linking international registries to FHIR and Phenopackets. NPJ Genom Med. 2025 Nov 18;10(1):72. doi: [10.1038/s41525-025-00534-z](https://doi.org/10.1038/s41525-025-00534-z). PMID: 41253795; PMCID: PMC12627670.
- [2] Graefe ASL, Hübner MR, **Rehburg F**, ... & Thun S. An ontology-based rare disease common data model harmonising international registries, FHIR, and Phenopackets. Sci Data. 2025 Feb 8;12(1):234. doi:[10.1038/s41597-025-04558-z](https://doi.org/10.1038/s41597-025-04558-z). PMID: 39922817; PMCID: PMC11807222.
- [3] Rekerle L, Danis D, **Rehburg F**, ... & Robinson PN. GA4GH phenopacket-driven characterization of genotype-phenotype correlations in Mendelian disorders. Am J Hum Genet. 2026 Jan 8;113(1):57-70. doi: [10.1016/j.ajhg.2025.12.001](https://doi.org/10.1016/j.ajhg.2025.12.001). Epub 2025 Dec 23. PMID: 41443197; PMCID: PMC12824607.
- [4] Danis D, Bamshad MJ, Bridges Y, ..., **Rehburg F**, ... & Robinson PN. A corpus of GA4GH phenopackets: Case-level phenotyping for genomic diagnostics and discovery. HGG Adv. 2025 Jan 9;6(1):100371. doi: [10.1016/j.xhgg.2024.100371](https://doi.org/10.1016/j.xhgg.2024.100371). Epub 2024 Oct 10. PMID: 39394689; PMCID: PMC11564936.

Conference

- [5] **Rehburg F**, Graefe A, Hübner M, Thun S. How Interoperability Can Enable Artificial Intelligence in Clinical Applications. Stud Health Technol Inform. 2024 Aug 22;316:596-600. doi: [10.3233/SHTI240485](https://doi.org/10.3233/SHTI240485). PMID: 39176813.
- [6] Graefe ASL, **Rehburg F**, Hübner M, Thun S, Beyan O. Review of Key Elements in Developing a Common Data Model for Rare Diseases: Identifying Common Success Factors. Stud Health Technol Inform. 2024 Aug 22;316:1396-1400. doi: [10.3233/SHTI240672](https://doi.org/10.3233/SHTI240672). PMID: 39176641.

Poster

- [7] **Rehburg, F**, Peeters, R, Wang, T, van Dam, P M., Bonizzi, P, & Karel, J (2023). Reconstruction of electrograms using a deep learning network trained on simulation data. Abstract from Electrocardiographic Imaging Summit 2023, Valencia, Spain. <https://cris.maastrichtuniversity.nl/en/publications/reconstruction-of-electrograms-using-a-deep-learning-network-trai/>